

Quantum Field Theory Past Paper Solutions

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The following is a collection of solutions to past exam questions for MAU44400-1 *Quantum Field Theory*, taught by prof S. Shatashvili. The (non-comprehensive) set of independent questions is

1. Scalar QED
2. $SO(N)$ invariant scalar field Theory
3. Lorentz transformation in Coulomb gauge
4. Four-point functions in $\frac{\lambda}{4!}\phi^4$ theory
5. 2D QED
6. Loop integrals
7. Massive gauge field
8. Maxwell-Chern-Simons theory
9. Current-current commutators in 2 space-time dimensions
10. Yukawa theory (**JM**)
11. Compton scattering (**JM**)
12. Radiative corrections (**JM**)
13. Supersymmetry
14. 2-loop in ϕ^4 theory
15. γ matrices
16. QED at tree level – Compton scattering
17. Nother current and massive electromagnetic field
18. Holomorphic quantisation
19. $e^-\mu \rightarrow e^-\mu^-$ tree level
20. QED two-loops
- 21.

1 Scalar QED

$$\begin{aligned}\mathcal{L} &= \mathcal{L}_{EM} + \mathcal{L}_{\text{mat}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + D^\mu\phi^*D_\mu\phi - m^2\phi^*\phi \\ \mathcal{L}_{EM} &= -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \\ \mathcal{L}_{\text{mat}} &= D^\mu\phi^*D_\mu\phi - m^2\phi^*\phi \\ D_\mu\phi &= (\partial_\mu + ieA_\mu)\phi \\ D_\mu\phi^* &= \partial_\mu\phi^* - ieA_\mu\phi^*.\end{aligned}$$

1.1 Part (a)

Write all equations of motion.

We see that $\mathcal{L} = \mathcal{L}(\phi, A, \partial\phi, \phi^*, \partial\phi^*, \partial A)$. The lagrangian is symmetric under exchange of $\phi \leftrightarrow \phi^*$, thus the equation of motion for one will be the same as the other up to this exchange.

$$\partial_k \frac{\partial \mathcal{L}}{\partial (\partial_k A_\mu)} - \frac{\partial \mathcal{L}}{\partial A_\mu} = 0 \quad (1.1)$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial A_\mu} &= \frac{\partial}{\partial A_\mu} D^\alpha \phi^* D_\alpha \phi = -ie\delta^{\alpha\mu}\phi^* D_\alpha \phi + ieD^\alpha \phi^* \phi \delta_\alpha^\mu \\ \frac{\partial \mathcal{L}}{\partial A_\mu} &= ie(\phi^* D_\mu \phi - D_\mu \phi^* \phi).\end{aligned}$$

and

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial (\partial_k A_\mu)} &= \frac{\partial}{\partial (\partial^k A_\mu)} \cdot \left(-\frac{1}{4} F^{\alpha\beta} F_{\alpha\beta} \right) = -\frac{1}{4} \cdot 2 \cdot F^{\alpha\beta} \cdot \frac{\partial F_{\alpha\beta}}{\partial (\partial_k A_\mu)} = -\frac{1}{2} F^{\alpha\beta} \cdot (\delta_\alpha^k \delta_\beta^\mu - \delta_\beta^k \delta_\alpha^\mu) \\ \frac{\partial \mathcal{L}}{\partial (\partial_k A_\mu)} &= -\frac{1}{2} [F^{k\mu} - F^{\mu k}] = F^{\mu k}.\end{aligned}$$

We get the equation of motion for A_k

$$\partial_k F^{\mu k} + ie(D_\mu \phi^* \phi - \phi^* D_\mu \phi) = 0 \quad (1.2)$$

equivalently

$$\partial_\mu F^{\mu k} + ie(\phi^* D_\mu \phi - D_\mu \phi^* \phi) = 0 \quad (1.3)$$

For ϕ 's equation of motion

$$\partial_k \frac{\partial \mathcal{L}}{\partial (\partial_k \phi_\mu)} - \frac{\partial \mathcal{L}}{\partial \phi_\mu} = 0 \quad (1.4)$$

$$\frac{\partial \mathcal{L}}{\partial \phi_\mu} = \frac{\partial}{\partial \phi_\mu} [D^\mu \phi^* D_\mu \phi - m^2 \phi^* \phi] = ieA_k D^k \phi^* - m^2 \phi^* \quad (1.5)$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_k \phi_\mu)} = D^\mu \phi^* \delta_\mu^k = D^k \phi^* \quad (1.6)$$

We have

$$\partial_k D^k \phi^* - ie A_k D^k \phi^* - m^2 \phi^* = 0 \quad (1.7)$$

Thus we get an equation of motion ϕ

$$D_k D^k \phi^* - m^2 \phi^* = 0 \quad (1.8)$$

and for ϕ^*

$$D_k D^k \phi - m^2 \phi = 0 \quad (1.9)$$

1.2 Part (b)

Write down the electric current $J^\mu = -\frac{\partial \mathcal{L}}{\partial A_\mu}$ in a manifestly gauge-invariant form and verify its conservation and verify its conservation $\partial_\mu J^\mu = 0$ We have

$$J^\mu = -\frac{\partial \mathcal{L}}{\partial A_\mu} = ie (\phi^* D_\mu \phi - D_\mu \phi^* \phi) \quad (1.10)$$

Notice it is unchanged under a gauge transform $\phi \rightarrow e^{i\alpha} \phi$

$$J^\mu \rightarrow J^{\mu'} = ie (\phi^* e^{-i\alpha} D_\mu e^{i\alpha} \phi - D_\mu \phi^* e^{-i\alpha} e^{i\alpha} \phi) = J^\mu \quad (1.11)$$

It forms a trivial $U(1)$ representation.

1.3 Part (c)

It would be useful to remember

$$T_{\text{mat}}^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} D^\nu \phi + \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} D^\nu \phi^* - \eta^{\mu\nu} \mathcal{L}_{\text{mat}} \quad (1.12)$$

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} = D^\mu \phi, \quad \frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^*)} = D^\mu \phi^*.$$

$$T_{\text{matt}}^{\mu\nu} = D^\mu \phi^* D^\nu \phi + D^\nu \phi^* D^\mu \phi - \eta^{\mu\nu} (D_\alpha \phi^* D^\alpha \phi - m^2 \phi^* \phi) \quad (1.13)$$

1.4 Part (d)

Use the scalar fields equations of motion and the non-commutativity of covariant derivatives

$$[D_\mu, D_\nu] \phi = ie F_{\mu\nu} \phi \quad [D_\mu, D_\nu] \phi^* = -ie F_{\mu\nu} \phi^* \quad (1.14)$$

to show

$$\partial_\mu T_{\text{matt}}^{\mu\nu} = F^{\nu\lambda} J_\lambda \quad (1.15)$$

Since $T^{\mu\nu}$ carries no charge and transforms trivially under $U(1)$, we have $\partial_\mu T^{\mu\nu} = D_\mu T^{\mu\nu}$.

$$\begin{aligned}
D_\mu T_{\text{matt}}^{\mu\nu} &= D_\mu [D^\mu \phi^*] D^\nu \phi + D^\mu \phi^* D_\mu [D^\nu \phi] + D_\mu [D^\nu \phi^*] D^\mu \phi + D^\nu \phi^* D_\mu [D^\mu \phi] \\
&\quad - D^\nu (D_\alpha \phi^* D^\alpha \phi - m^2 \phi^* \phi) \\
&= D_\mu [D^\mu \phi^*] D^\nu \phi + D^\mu \phi^* D_\mu [D^\nu \phi] + D_\mu [D^\nu \phi^*] D^\mu \phi + D^\nu \phi^* D_\mu [D^\mu \phi] \\
&\quad - D^\nu [D_\alpha \phi^*] D^\alpha \phi - D_\alpha \phi^* D^\nu [D^\alpha \phi] - m^2 D^\nu \phi^* \phi - m^2 \phi^* D^\nu \phi
\end{aligned}$$

$$\begin{aligned}
&= m^2 \phi^* D^\nu \phi + D^\mu \phi^* D_\mu [D^\nu \phi] + m^2 \phi^* D^\mu \phi + D^\nu \phi^* D_\mu [D^\mu \phi] \\
&\quad - D^\nu [D_\alpha \phi^*] D^\alpha \phi - D_\alpha \phi^* D^\nu [D^\alpha \phi] - m^2 D^\nu \phi^* \phi - m^2 \phi^* D^\nu \phi \\
&= D^\mu \phi^* D_\mu [D^\nu \phi] + D^\nu \phi^* D_\mu [D^\mu \phi] - D^\nu [D_\alpha \phi^*] D^\alpha \phi + D_\alpha \phi^* D^\nu [D^\alpha \phi]
\end{aligned}$$

$\alpha \rightarrow \mu$

$$= D^\mu \phi^* D_\mu [D^\nu \phi] + D^\nu \phi^* D_\mu [D^\mu \phi] - D^\nu [D_\mu \phi^*] D^\mu \phi - D_\mu \phi^* D^\nu [D^\mu \phi]$$

First and last terms combine using the first relation, second and third combine using the second relation, these differ by neg thus we get the stress to be conserved.

2 2025

2.1 Question 6: Loop Integrals

a) 5 marks: Show that for $m > 2$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{[l_E^2 + \Delta]^m} = \frac{1}{(4\pi)^2} \frac{1}{(m-1)(m-2)} \frac{1}{\Delta^{m-2}}$$

b) 5 marks Show that for $m > 3$:

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{l^2}{[l_E^2 + \Delta]^m} = \frac{1}{(4\pi)^2} \frac{1}{(m-1)(m-2)(m-3)} \frac{1}{\Delta^{m-3}}$$

c) 10 marks: Prove that for $n \geq 2$:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{[l_E^2 + \Delta]^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n - \frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n - \frac{d}{2}}$$

d) 5 marks: Prove that for $n \geq 2$:

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{l^2}{[l_E^2 + \Delta]^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n - \frac{d}{2} - 1)}{\Gamma(n)} \left(\frac{1}{\Delta}\right)^{n - \frac{d}{2} - 1}$$

Figure 2.1

Firstly there is a typo in **part d**¹ there should be a factor of $\frac{d}{2}$ on the rhs. We notice that **part a** is a special case of **part c** and **part b** is a special case of **part d**. Thus we may solve the general case then specify the required conditions to verify the special cases.

As you know, the surface area of the unit sphere S^{d-1} in d Euclidean dimensions is

$$\Omega_{d-1} = \int d\Omega_{d-1} = \frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})} \quad (2.1)$$

2.1.1 Part c

We can consider a function of l_E^2 only we can use d -dim spherical coords.

$$\int \frac{d^d l_E}{(2\pi)^d} f(l_E^2) = \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty l^{d-1} f(l^2) dl \quad (2.2)$$

Let $x = l^2$, $dx = 2l dl$, thus, $l^{d-1} dl = \frac{1}{2} x^{\frac{d}{2}-1} dx$. One can define $a := \frac{d}{2}$ and $f(l^2) := \frac{1}{(l^2 + \Delta)^n}$.

$$\frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty l^{d-1} \frac{1}{(l^2 + \Delta)^n} dl = \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{1}{2} x^{a-1} \frac{1}{(x + \Delta)^n} dx$$

¹ $\exists$ another 1 ?

We notice the form of a β function integral

$$\int_0^\infty \frac{x^{a-1}}{(x+\Delta)^c} dx = \Delta^{a-c} \frac{\Gamma(a) \Gamma(c-a)}{\Gamma(c)} \quad (2.3)$$

$$\begin{aligned} \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{1}{2} x^{d-1} \frac{1}{(x+\Delta)^n} dx &= \frac{\Omega_{d-1}}{(2\pi)^d} \cdot \Delta^{\frac{d}{2}-n} \frac{\Gamma(\frac{d}{2}) \Gamma(n-\frac{d}{2})}{\Gamma(n)} \\ \frac{\Omega_{d-1}}{(2\pi)^d} \cdot \Delta^{\frac{d}{2}-n} \frac{\Gamma(\frac{d}{2}) \Gamma(n-\frac{d}{2})}{\Gamma(n)} &= \frac{\frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}}{(2\pi)^d} \cdot \Delta^{\frac{d}{2}-n} \frac{\Gamma(\frac{d}{2}) \Gamma(n-\frac{d}{2})}{\Gamma(n)} \\ \frac{\frac{2\pi^{\frac{d}{2}}}{\Gamma(\frac{d}{2})}}{(2\pi)^d} \cdot \Delta^{\frac{d}{2}-n} \frac{\Gamma(\frac{d}{2}) \Gamma(n-\frac{d}{2})}{\Gamma(n)} &= \frac{\Delta^{\frac{d}{2}-n}}{(4\pi)^{\frac{d}{2}}} \cdot \frac{\Gamma(n-\frac{d}{2})}{\Gamma(n)}. \end{aligned}$$

Thus we have verified **c**, namely that

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{1}{(l_E^2 + \Delta)^n} = \frac{1}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(n-\frac{d}{2})}{\Gamma(n)} \left(\frac{1}{\Delta^{n-\frac{d}{2}}} \right) \quad \forall n \geq 2 \quad (2.4)$$

The requirement on n follows from the necessity of convergence of (2.4) for $l_E \rightarrow \infty$.

2.1.2 Part a

Notice that **part a** is the same as **part c** but with $d = 4$. Thus, we let $d = 4$.

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{(l_E^2 + \Delta)^n} = \frac{1}{(4\pi)^2} \frac{\Gamma(n-2)}{\Gamma(n)} \left(\frac{1}{\Delta^{n-2}} \right).$$

Now we use the property of Γ functions, $\Gamma(z) = (z-1) \Gamma(z)$.

$$\frac{\Gamma(n-2)}{\Gamma(n)} = \frac{\Gamma(n-2)}{(n-1) \Gamma(n-1)} = \frac{\Gamma(n-2)}{(n-1)(n-2) \Gamma(n-2)} = \frac{1}{(n-1)(n-2)} \quad (2.5)$$

Substituting we have shown verified **part a**.

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{1}{(l_E^2 + \Delta)^n} = \frac{1}{(4\pi)^2} \frac{1}{(n-1)(n-2)} \left(\frac{1}{\Delta^{n-2}} \right) \quad \forall n \geq 2 \quad (2.6)$$

2.1.3 Part d

We notice that we have an extra factor of l_E^2 compared to **part c**. Let $f(l^2) := \frac{l^2}{(l^2 + \Delta)^n}$

$$\begin{aligned} \int \frac{d^d l_E}{(2\pi)^d} f(l_E^2) &= \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty l^{d-1} f(l^2) dl \\ \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty l^{d-1} f(l^2) dl &= \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{l^{d+1}}{(l^2 + \Delta)^n} dl \\ \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{l^{d+1}}{(l^2 + \Delta)^n} dl &= \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{x^{\frac{d}{2}+1}}{(x + \Delta)^n} dx \\ \frac{\Omega_{d-1}}{(2\pi)^d} \int_0^\infty \frac{x^{\frac{d}{2}+1}}{(x + \Delta)^n} dx &= \frac{\Omega_{d-1}}{(2\pi)^d} \Delta^{\frac{d}{2}-n+1} \Gamma\left(\frac{d}{2} + 1\right) \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)} \end{aligned}$$

We can use the property $\Gamma(z+1) = z\Gamma(z)$ ($z = \frac{d}{2}$ here).

$$\begin{aligned} \frac{\Omega_{d-1}}{(2\pi)^d} \Delta^{\frac{d}{2}-n+1} \Gamma\left(\frac{d}{2} + 1\right) \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)} &= \frac{d}{2} \frac{\Omega_{d-1}}{(2\pi)^d} \Delta^{\frac{d}{2}-n+1} \Gamma\left(\frac{d}{2}\right) \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)} \\ \frac{d}{2} \frac{\left(\frac{2\pi^{\frac{d}{2}}}{\Gamma\left(\frac{d}{2}\right)}\right)}{(2\pi)^d} \Delta^{\frac{d}{2}-n+1} \Gamma\left(\frac{d}{2}\right) \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)} &= \frac{d}{2} \frac{1}{(4\pi)^{\frac{d}{2}}} \Delta^{\frac{d}{2}-n+1} \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)}. \end{aligned}$$

Thus we have verified **part d**

$$\int \frac{d^d l_E}{(2\pi)^d} \frac{l_E^2}{(l_E^2 + \Delta)^n} dl_E = \frac{d}{2} \frac{1}{(4\pi)^{\frac{d}{2}}} \Delta^{\frac{d}{2}-n+1} \frac{\Gamma\left(n - \frac{d}{2} + 1\right)}{\Gamma(n)} \quad (2.7)$$

2

2.1.4 Part b

Let $d = 4$ in (2.7).

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{l_E^2}{(l_E^2 + \Delta)^n} dl_E = 2 \frac{1}{(4\pi)^2} \Delta^{3-n} \frac{\Gamma(n-3)}{\Gamma(n)} \quad (2.8)$$

$$\begin{aligned} \frac{\Gamma(n-3)}{\Gamma(n)} &= \frac{\Gamma(n-3)}{(n-1)\Gamma(n-1)} = \frac{\Gamma(n-3)}{(n-1)(n-2)\Gamma(n-2)} \\ \frac{\Gamma(n-3)}{(n-3)(n-2)(n-1)\Gamma(n-3)} &= \frac{1}{(n-3)(n-2)(n-1)}. \end{aligned}$$

$$\int \frac{d^4 l_E}{(2\pi)^4} \frac{l_E^2}{(l_E^2 + \Delta)^n} dl_E = \frac{2}{(4\pi)^2} \frac{1}{(n-3)(n-2)(n-1)} \frac{1}{\Delta^{n-3}} \quad (2.9)$$

²prob a -1 in the gamma

3 2024

3.1 Question 1: The Massive Gauge Field

Given the action for a massive gauge field

$$A^\mu(\phi, \partial_\mu \phi) = \int d^4x \left[-\frac{1}{2} (\partial_\mu \phi^\nu)^2 + \frac{1}{2} (\partial_\mu \phi^\mu)^2 + \frac{m^2}{2} \phi_\mu \phi^\mu \right].$$

3.1.1 (a) Derive the equations of motion.

The equations of motion are given by

$$\partial_\alpha \frac{\partial \mathcal{L}}{\partial (\partial_\alpha \phi^\beta)} - \frac{\partial \mathcal{L}}{\partial \phi^\beta} = 0 \quad (3.1)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \phi^\beta} &= \frac{\partial}{\partial \phi^\beta} \left[\frac{m^2}{2} \phi_\mu \phi^\mu \right] \\ &= m^2 \phi_\mu \delta_\beta^\mu \\ &= m^2 \phi_\beta. \end{aligned}$$

$$\begin{aligned} \partial_\alpha \frac{\partial \mathcal{L}}{\partial (\partial_\alpha \phi^\beta)} &= \partial_\alpha \frac{\partial}{\partial (\partial_\alpha \phi^\beta)} \left[-\frac{1}{2} (\partial_\mu \phi^\nu)^2 + \frac{1}{2} (\partial_\mu \phi^\mu)^2 \right] \\ &= \partial_\alpha \left[-(\partial_\mu \phi^\nu) \delta_\mu^\alpha \delta_\beta^\nu + (\partial_\gamma \phi^\gamma) \delta_\mu^\alpha \delta_\beta^\mu \right] \\ &= \partial_\alpha \left[-(\partial^\alpha \phi_\beta) + (\partial_\gamma \phi^\gamma) \delta_\beta^\alpha \right] \\ &= -\partial_\alpha \partial^\alpha \phi_\beta + \partial_\beta \partial_\gamma \phi^\gamma \\ &= -\partial_\alpha \partial^\alpha \phi_\beta + \partial_\beta \partial_\alpha \phi^\alpha \\ &= \partial_\alpha F_\beta^\alpha. \end{aligned}$$

Thus, we have,

$$\begin{aligned} \partial_\alpha \frac{\partial \mathcal{L}}{\partial (\partial_\alpha \phi^\beta)} - \frac{\partial \mathcal{L}}{\partial \phi^\beta} &= 0 \\ \partial_\alpha F_\beta^\alpha - m^2 \phi_\beta &= 0. \end{aligned}$$

3.1.2 (b) Show that these equations imply that $\partial_\mu \phi^\mu = 0$.

In order to show the relation is satisfied we use the general property of tensors. Namely, the contraction of an anti-symmetric tensor with a symmetric tensors vanishes.

$$\begin{aligned} \partial^\beta \partial_\alpha F_\beta^\alpha - m^2 \partial^\beta \phi_\beta &= 0 \\ -m^2 \partial^\mu \phi_\mu &= 0 \\ \partial_\mu \phi^\mu &= 0. \end{aligned}$$

3.1.3 (c) Construct the Hamiltonian density, and constraints if there are such, as functionals of ϕ_μ , $\partial_i \phi_\mu$ and the generalised momentum Π_μ .

The Hamiltonian Density is given by

$$\mathcal{H} = \Pi_\mu (\partial_0 \phi)^\mu - \mathcal{L} \quad (3.2)$$

where $\Pi_\mu := \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^\mu)}$.

By inspection, our Lagrangian density is

$$\mathcal{L} = -\frac{1}{4} F^{\mu\nu} F_{\mu\nu} + \frac{m^2}{2} \phi_\mu \phi^\mu \quad (3.3)$$

If we use the convention of Greek indices run from 0, 1, 2, 3, and Latin indices run from 1, 2, 3.

$$\mathcal{L} = -\frac{1}{4} [F_{0i} F^{0i} + F_{i0} F^{i0} + F_{ij} F^{ij}] + \frac{m^2}{2} \phi_\mu \phi^\mu \quad (3.4)$$

We can use the mostly negative signature, and notice that since F_{0i} and F^{0i} are totally anti-symmetric their contraction is symmetric, $F_{0i} F^{i0} = F_{i0} F^{i0}$

$$\mathcal{L} = -\frac{1}{2} F_{0i} F^{0i} - \frac{1}{4} F^{ij} F_{ij} + \frac{m^2}{2} (\phi_0^2 - \phi_i \phi^i) \quad (3.5)$$

Since we have no time derivative on the temporal component of ϕ , we have our first constraint

$$\mathcal{C}_1 = \Pi_0 = 0 \quad (3.6)$$

If we look at the spatial momenta (0+3 Euclidean space)

$$\begin{aligned} \Pi_i &= \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi^i)} \\ &= -\frac{1}{2} \frac{\partial}{\partial(\partial_0 \phi^i)} [(\partial_0 \phi_j - \partial_j \phi_0) (\partial^0 \phi^j - \partial^j \phi^0)] \\ &= -\frac{1}{2} [\delta_{ij} (\partial^0 \phi^j - \partial^j \phi^0) + (\partial_0 \phi_j - \partial_j \phi_0) \delta_i^j] \\ &= -F_{0i}. \end{aligned}$$

Thus we have the expression, which is our second constraint $\mathcal{C}_2$

$$\dot{\phi}^i = \partial_i \phi_0 - \Pi_i \quad (3.7)$$

We had the expression for the Canonical Hamiltonian density

$$\mathcal{H}_C = \Pi_\mu \dot{\phi}^\mu - \mathcal{L} \quad (3.8)$$

The two constraints imply

$$\mathcal{H}_c = \Pi_i (\Pi_i - \partial_i \phi_0) - \mathcal{L} \quad (3.9)$$

Notice that we found $F_{0i} F^{0i} = \Pi_i \Pi_i$. Consequently, expanding the Lagrangian density gives

$$\mathcal{H}_C = \Pi_i \Pi_i - \Pi_i \partial_i \phi_0 - \frac{1}{2} \Pi_i \Pi_i + \frac{1}{4} F_{ij} F^{ij} - \frac{m^2}{2} \phi_0^2 + \frac{m^2}{2} \phi_i \phi_i \quad (3.10)$$

$$\mathcal{H}_C = \frac{1}{2}\Pi_i\Pi_i - \Pi_i\partial_i\phi_0 + \frac{1}{4}F_{ij}F^{ij} - \frac{m^2}{2}\phi_0^2 + \frac{m^2}{2}\phi_i\phi_i \quad (3.11)$$

The second term can be expressed as

$$-\Pi_i\partial_i\phi_0 = \phi_0\partial_i\Pi_i - \partial_i(\phi_0\Pi_i) \quad (3.12)$$

where we can drop the second term of this reformulation as it is a boundary term.

$$\mathcal{H}_C = \frac{1}{2}\Pi_i\Pi_i + \phi_0\partial_i\Pi_i + \frac{1}{4}F_{ij}F^{ij} - \frac{m^2}{2}\phi_0^2 + \frac{m^2}{2}\phi_i\phi_i \quad (3.13)$$

The total Hamiltonian density includes the primary constraint $\mathcal{H}_T = \mathcal{H}_C + U(x)\Pi_0$. The first constraint must be preserved in time, thus after applying the constraints we have

$$\{\Pi_0, \mathcal{H}_T\} = 0 \quad (3.14)$$

The multiplier term doesn't contribute so we have $\{\Pi_0, \mathcal{H}_T\} = \{\Pi_0, \mathcal{H}_C\}$. Using the Canonical Poisson Bracket $\{\phi_\mu(x), \Pi^\nu(y)\} = \delta^{(3)}(x-y)$.

In general if we have two functionals

$$\{F[\phi, \Pi], G[\phi, \Pi]\} = \int d^3y \left[\frac{\delta F}{\delta \phi_\mu(y)} \frac{\delta G}{\delta \Pi^\mu(y)} - \frac{\delta F}{\delta \Pi^\mu(y)} \frac{\delta G}{\delta \phi_\mu(y)} \right] \quad (3.15)$$

Let $F = \Pi^0(x)$ and $G = \mathcal{H}_C$. Then we can use

$$\frac{\delta \Pi^0(x)}{\delta \phi_\mu(x)} = 0 \quad \frac{\delta \Pi^0(x)}{\delta \Pi^\mu(y)} = \delta_\mu^0 \delta^{(3)}(x-y) \quad (3.16)$$

$$\begin{aligned} \{\Pi^0(x), \mathcal{H}_C\} &= - \int d^3y \delta_\mu^0 \delta^{(3)}(x-y) \frac{\delta \mathcal{H}_C}{\delta \phi_\mu(y)} \\ \{\Pi^0(x), \mathcal{H}_C\} &= - \frac{\delta \mathcal{H}_C}{\delta \phi_0(x)}. \end{aligned}$$

where, ¹

$$\begin{aligned} -\frac{\delta \mathcal{H}_C}{\delta \phi_0(x)} &= m^2\phi_0 - \partial_i\Pi_i \\ \dot{\Pi}^0 &= -\frac{\delta \mathcal{H}_C}{\delta \phi_0(x)} = m^2\phi_0 - \partial_i\Pi_i \end{aligned}$$

Demanding that $\dot{\Pi} = 0$, and that it holds for evolution $\dot{\Pi} \approx 0$. So our first and second constraints are

$$\begin{aligned} \mathcal{C}_1 &= \Pi^0 = 0 \\ \mathcal{C}_2 &= m^2\phi_0 - \partial_i\Pi_i = 0. \end{aligned}$$

¹check - sign

$$\{\mathcal{C}_1, \mathcal{C}_2\} = -m^2 \delta^{(3)}(x - y) \quad (3.17)$$

Thus even if we set $\mathcal{C}_1$ or $\mathcal{C}_2$ to zero. The Poisson-bracket does not vanish. We conclude that the constraints are second class. A second-class constraint removes phase-space variables. Unlike a first class constraint it does not represent a gauge redundancy.

$$\mathcal{C}_2 = 0 \quad \implies \quad \phi_0 = \frac{1}{m^2} \partial_i \Pi_i \quad (3.18)$$

If we substitute this in we can remove the dependence on ϕ_0 and obtain the reduced Hamiltonian

$$\mathcal{H}_R = \frac{1}{2} \Pi_i \Pi_i + \frac{1}{2m^2} (\Pi_i \Pi_i)^2 + \frac{1}{4} F_{ij} F^{ij} + \frac{m^2}{2} \phi_i \phi_i \quad (3.19)$$

This form makes it apparent that the massive gauge field only has three degree of freedom which correspond to the possible polarisations.

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$$A = \int d^4x \left[\partial_\mu \phi^* \partial^\mu \phi + i \chi^+ \bar{\sigma} \cdot \partial \chi + F^* F \right] \quad (4.1)$$

Note $\chi^+ := \chi^\dagger$ (convention).

4.1 (a)

4.2 (b)

Given the new infinitesimal transformations

$$\begin{aligned} \delta \phi &= -i \varepsilon^T \sigma^2 \chi \\ \delta \chi &= \varepsilon F - \sigma \cdot \partial \phi \sigma^2 \varepsilon^* \\ \delta F &= -i \varepsilon^+ \bar{\sigma} \cdot \partial \chi. \end{aligned}$$

¹ We need to calculate $\delta \phi^*$, $\delta \chi^*$ and δF^* .

$$\begin{aligned} \delta \phi^* &= -i \chi^* \sigma^2 \varepsilon^* \\ \delta \chi^\dagger &= F^* \varepsilon^\dagger - \varepsilon^T \sigma^2 \sigma \cdot \partial \phi^* \\ \delta F^* &= i (\partial_\mu \chi^*) \bar{\sigma}^\mu \varepsilon. \end{aligned}$$

where we used the properties

$$(\sigma^2)^* = -\sigma^2, \quad (\sigma^2)^T = -\sigma^2 \quad (4.2)$$

Thus we can vary our action

$$\begin{aligned} \delta A &= \int d^4x \left[\delta (\partial_\mu \phi^* \partial^\mu \phi) + \delta (i \chi^+ \bar{\sigma} \cdot \partial \chi) + \delta (F^* F) \right] \\ &= \int d^4x \left[(\partial_\mu \delta \phi^*) \partial^\mu \phi + (\partial_\mu \phi^*) (\partial^\mu \delta \phi) + i (\delta \chi^+ \bar{\sigma}) \cdot \partial \chi + i \chi^+ \bar{\sigma} \cdot (\partial \delta \chi) + (\delta F^*) F + F^* (\delta F) \right] \end{aligned}$$

First contribution C_1 gives

$$\begin{aligned} (\partial_\mu \delta \phi^*) \partial^\mu \phi + (\partial_\mu \phi^*) (\partial^\mu \delta \phi) &= (-i \partial_\mu \chi^* \sigma^2 \varepsilon^*) \partial^\mu \phi - \partial_\mu \phi^* (i \varepsilon^T \sigma^2 \partial^\mu \chi) \\ &= i \left[-\partial_\mu \chi^* \sigma^2 \varepsilon^* \partial^\mu \phi - \partial_\mu \phi^* \varepsilon^T \sigma^2 \partial^\mu \chi \right]. \end{aligned}$$

Second contribution C_2 gives

$$\begin{aligned} i (\delta \chi^+ \bar{\sigma}^\nu) \cdot \partial_\nu \chi + i \chi^+ \bar{\sigma}^\nu \cdot (\partial_\nu \delta \chi) &= i \left(\left(F^* \varepsilon^\dagger - \varepsilon^T \sigma^2 \sigma \cdot \partial \phi^* \right) \bar{\sigma}^\nu \right) \partial_\nu \chi + i \chi^+ \bar{\sigma}^\nu (\partial_\nu (\varepsilon F - \sigma \cdot \partial \phi \sigma^2 \varepsilon^*)) \\ &= i \left[F^* \varepsilon^\dagger \bar{\sigma}^\nu \partial_\nu \chi - \varepsilon^T \sigma^2 \sigma^\alpha \cdot \partial_\alpha \phi^* \bar{\sigma}^\nu \partial_\nu \chi + \chi^+ \bar{\sigma}^\nu \varepsilon \partial_\nu F - \chi^+ \bar{\sigma}^\nu \partial_\nu \sigma^\alpha \cdot \partial_\alpha \phi \sigma^2 \varepsilon^* \right] \end{aligned}$$

¹Note σ^2 is the 2nd Pauli matrix, not a scalar

The third contribution C_3 gives

$$\begin{aligned} (\delta F^*) F + F^* (\delta F) &= (i (\partial_\mu \chi^*) \bar{\sigma}^\mu \varepsilon) F + F^* (-i \varepsilon^+ \bar{\sigma} \cdot \partial \chi) \\ &= i [\partial_\mu \chi^* \bar{\sigma}^\mu \varepsilon F - F^* \varepsilon^+ \bar{\sigma}^\mu \partial_\mu \chi]. \end{aligned}$$

Notice the first term in C_2 cancels with the second term in C_3 . Combining things and letting go of the overall i : We define $iC = C_1 + C_2 + C_3$

$$\begin{aligned} C &= -\partial_\mu \chi^* \sigma^2 \varepsilon^* \partial^\mu \phi - \partial_\mu \phi^* \varepsilon^T \sigma^2 \partial^\mu \chi - \varepsilon^T \sigma^2 \sigma^\alpha \partial_\alpha \phi^* \bar{\sigma}^\nu \partial_\nu \chi + \chi^+ \bar{\sigma}^\nu \varepsilon \partial_\nu F - \chi^+ \bar{\sigma}^\nu \partial_\nu \sigma^\alpha \cdot \partial_\alpha \phi \sigma^2 \varepsilon^* + \partial_\mu \chi^* \bar{\sigma}^\mu \varepsilon F \\ C &= -\partial_\mu \chi^* \sigma^2 \varepsilon^* \partial^\mu \phi - \chi^+ \bar{\sigma}^\nu \partial_\nu \sigma^\alpha \partial_\alpha \phi \sigma^2 \varepsilon^* - \partial_\mu \phi^* \varepsilon^T \sigma^2 \partial^\mu \chi - \varepsilon^T \sigma^2 \sigma^\alpha \partial_\alpha \phi^* \bar{\sigma}^\nu \partial_\nu \chi + \chi^+ \bar{\sigma}^\nu \varepsilon \partial_\nu F + \partial_\nu \chi^* \bar{\sigma}^\nu \varepsilon F \\ C &= \sigma^2 \varepsilon^* [-\partial_\mu \chi^* \partial^\mu \phi - \chi^+ \bar{\sigma}^\nu \sigma^\alpha \partial_\nu \partial_\alpha \phi] - \varepsilon^T \sigma^2 [\partial_\mu \phi^* \partial^\mu \chi + \sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \phi^* \partial_\nu \chi] + \cancel{\partial_\nu (\chi^+ \bar{\sigma}^\nu \varepsilon F)} \\ &\cdot \end{aligned}$$

For the $-\varepsilon^T \sigma^2 \sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \phi^* \partial_\nu \chi$ term we can use $u dv = d(uv) - v du$

$$-\varepsilon^T \sigma^2 \sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \phi^* \partial_\nu \chi = -\partial_\nu [\varepsilon^T \sigma^2 \sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \phi^* \chi] + \varepsilon^T \sigma^2 \sigma^\alpha \bar{\sigma}^\nu \chi \partial_\nu \partial_\alpha \phi^*.$$

Likewise,

$$-\varepsilon^T \sigma^2 \partial_\mu \phi^* \partial^\mu \chi = -\partial_\mu [\varepsilon^T \sigma^2 \partial_\mu \phi^* \chi] + \varepsilon^T \sigma^2 \partial^\mu \partial_\mu \phi^* \chi.$$

Using $\sigma^\alpha \bar{\sigma}^\nu + \sigma^\nu \bar{\sigma}^\alpha = 2\eta^{\alpha\nu} \mathbf{1}_{2 \times 2}$, so terms with $\sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \partial_\nu \phi = \frac{1}{2} [\sigma^\alpha \bar{\sigma}^\nu + \bar{\sigma}^\alpha \sigma^\nu] \partial_\nu \partial_\alpha \phi = \eta^{\alpha\nu} \partial_\nu \partial_\alpha \phi = \square \phi$. We notice for the second term only the symmetric term in this identity survives since the product of a symmetric tensor and anti-symmetric tensor is zero.

If we perform the same calculation for the ε^* terms we get the result

$$\begin{aligned} C &= -\partial_\nu [\varepsilon^* \sigma^2 \chi^\dagger \partial_\alpha \phi] + \varepsilon^* \sigma^2 \chi^\dagger \partial_\mu \partial^\mu \phi - \varepsilon^* \sigma^2 \chi^\dagger \partial_\nu \partial_\alpha \phi - \partial_\mu [\varepsilon^T \sigma^2 \partial_\mu \phi^* \chi] + \varepsilon^T \sigma^2 \square \phi^* \chi \\ &\quad - \partial_\nu [\varepsilon^T \sigma^2 \sigma^\alpha \bar{\sigma}^\nu \partial_\alpha \phi^* \chi] + \varepsilon^T \sigma^2 \chi \square \phi^* \\ &\cdot \end{aligned}$$

XXX WRONG

5 Question 4: Four-point functions in $\frac{\lambda}{4!}\phi^4$ theory

Draw all diagrams with four external legs in ϕ^4 theory up to one loop (there is one diagram in zero loop and three diagrams in one loop.

$$i\mathcal{M}(p_1 p_2 \rightarrow p_3 p_4) = -i\lambda + (-i\lambda)^2 [iV(s) + iV(t) + iV(u)] \quad (5.1)$$

Figure 5.1: Tree level Diagram for 4pt functions in ϕ^4 theory.

Figure 5.2: S channel

Figure 5.3: t channel

For one loop we expand the interaction to second order, $\frac{1}{2!} \left(\frac{-i\lambda}{4!}\right)^2$. Thus, there exists two interaction vertices, call them z and w . Each have fields $\phi^4(z)$ and $\phi^4(w)$, so there are eight identical fields since there exists four external fields ϕ_i , $i = 1, 2, 3, 4$. Take ϕ_1 , it can contract with any of the ϕ 's. For example, say it contracts with one of the z fields. Thus, there exists three remaining z fields to choose from.

For the s channel ϕ_1 and ϕ_2 attach to the same vertex, thus if ϕ_1 picks z , then ϕ_2 must also. The two remaining ϕ 's corresponding to z will be internal.

Now ϕ_3 must attach to the other vertex with four choices, ϕ_4 attaches to one of the three remaining (the two remaining for w are internal and form the loop with the remaining two of z). This loop can be formed by: At z take one of the two remaining fields ϕ and contract with one of the two remaining at w .

Thus we chose the one from 8 possible fields, one from the 3 z remaining fields, one from the 4 w fields, another from the 3 remaining w fields, and 2 possible internal fields of w . This gives us a factor of $8 \cdot 3 \cdot 4 \cdot 3 \cdot 2 = 576$.

From our expansion we had $\frac{1}{2!} (-i\lambda)^2 \cdot \left(\frac{1}{4!}\right)^2 = \frac{1}{2!} (-i\lambda)^2 \cdot \frac{1}{576}$ taking the product we get the net symmetry factor of $\frac{1}{2}$ which can be absorbed into the definition of our V (madelstrams).

5.1 Part (a)

Since we have

$$V(S) = \frac{1}{2} \int d^4k \frac{1}{k^2 - m^2 - i\epsilon} \frac{1}{(k+P)^2 - m^2 - i\epsilon} \quad (5.2)$$

We can use the Feynman parameter,

$$\frac{1}{AB} = \int_0^1 dx \frac{1}{[xA + (1-x)B]^2} \quad (5.3)$$

Figure 5.4: u channel

Define $A := k^2 - m^2 - i\varepsilon$, $B := (k - P)^2 - m^2 - i\varepsilon$. Then we have

$$V(S) = \frac{1}{2} \int d^4k \int_0^1 dx \frac{1}{\left[x(k^2 - m^2 - i\varepsilon) + (1-x) \left((k-P)^2 - m^2 - i\varepsilon \right) \right]^2} \quad (5.4)$$

Looking at what is inside the square bracket:

$$k^2 - m^2 - i\varepsilon + (1-x) 2k \cdot P + (1-x) P^2.$$

Completing the square,

$$k^2 + (1-x) 2k \cdot P = [k + (1-x) P]^2 - (1-x)^2 P^2 \quad (5.5)$$

So we get

$$[k + (1-x) P]^2 - (1-x)^2 P^2 + (1-x) P^2 - m^2 - i\varepsilon \quad (5.6)$$

$$\begin{aligned} & [k + (1-x) P]^2 - (1-x)^2 P^2 + (1-x) P^2 - m^2 - i\varepsilon \\ &= [k + (1-x) P]^2 + (1-x) [1 - (1-x)] P^2 - m^2 - i\varepsilon \\ &= [k + (1-x) P]^2 + (1-x) x P^2 - m^2 - i\varepsilon \end{aligned}$$

Define the shifted momentum, $\ell = k + (1-x) P$ and $\Delta := (1-x) x P^2 - m^2$

$$V(S) = \frac{1}{2} \int d^4k \int_0^1 dx \frac{1}{[\ell^2 + \Delta - i\varepsilon]^2} \quad (5.7)$$

5.2 Part (b)

Use dimensional regularization to evaluate momentum integral (you dont have to integrate over Feynman parameter)

We can use the dimensional regularisation formula

$$\int \frac{d^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i\varepsilon)^\alpha} = \frac{i(-1)^\alpha}{(4\pi)^{\frac{d}{2}}} \frac{\Gamma(\alpha - \frac{d}{2})}{\Gamma(\alpha)} (\Delta - i\varepsilon)^{\frac{d}{2} - \alpha} \quad (5.8)$$

In our case: $\alpha = 2$, $(-1)^2 = 1$, $\Gamma(2) = 1$ gives

$$iV(S) = -\frac{1}{2} \mu^{2\varepsilon} \int_0^1 dx \frac{i}{(4\pi)^{\frac{d}{2}}} \Gamma\left(2 - \frac{d}{2}\right) (\Delta - i\varepsilon)^{\frac{d}{2} - 2} \quad (5.9)$$

We can set $d = 4 - 2\varepsilon$

$$iV(S) = -\frac{1}{2} \frac{1}{16\pi^2} \left(\frac{\mu^2}{\Delta - i\varepsilon} \right)^\varepsilon \int_0^1 dx \frac{i}{(4\pi)^{-\varepsilon}} \Gamma(\varepsilon) \quad (5.10)$$

We can taylor expand

$$\left(\frac{\mu^2}{\Delta - i\varepsilon}\right)^\varepsilon = 1 + \varepsilon \ln\left(\frac{\mu^2}{\Delta - i\varepsilon}\right) + \mathcal{O}(\varepsilon^2) \quad (5.11)$$

$$\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_E + \mathcal{O}(\varepsilon) \quad (5.12)$$

$$(4\pi)^\varepsilon = 1 + \varepsilon \ln(4\pi) + \mathcal{O}(\varepsilon^2) \quad (5.13)$$

where γ_E is the Euler macharoni constant

$$iV(S) = -\frac{i}{32\pi^2} \left(1 + \varepsilon \ln\left(\frac{\mu^2}{\Delta - i\varepsilon}\right)\right) \int_0^1 dx (1 + \varepsilon \ln(4\pi)) \left(\frac{1}{\varepsilon} - \gamma_E\right) \quad (5.14)$$

Up to order $\mathcal{O}(\varepsilon^0)$

$$iV(S) = -\frac{i}{32\pi^2} \int_0^1 dx \left[\frac{1}{\varepsilon} - \gamma_E + \ln 4\pi + \ln\left(\frac{\mu^2}{\Delta - i\varepsilon}\right) \right] \quad (5.15)$$

6 Question 2: Gamma Matrices

Assume

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2\eta^{\mu\nu} \quad (6.1)$$

and the definition

$$S^{\mu\nu} = -S^{\nu\mu} = \frac{1}{4i} [\gamma^\mu, \gamma^\nu] \quad (6.2)$$

but do not assume the specific form of the 4×4 matrices γ^μ and $S^{\mu\nu}$ Part (a) Show

$$[S^{\kappa\lambda}, \gamma^\mu] = -i\eta^{\lambda\mu}\gamma^\kappa + i\eta^{\kappa\mu}\gamma^\lambda \quad (6.3)$$

We have

$$[S^{\kappa\lambda}, \gamma^\mu] = \left[\frac{1}{4i} [\gamma^\kappa, \gamma^\lambda], \gamma^\mu \right] \quad (6.4)$$

$$\begin{aligned} [S^{\kappa\lambda}, \gamma^\mu] &= \frac{1}{4i} (\gamma^\kappa \gamma^\lambda \gamma^\mu - \gamma^\lambda \gamma^\kappa \gamma^\mu - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa) \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= -\gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\kappa \gamma^\lambda \gamma^\mu + \gamma^\lambda \gamma^\mu \gamma^\kappa - \gamma^\lambda \gamma^\mu \gamma^\kappa - \gamma^\lambda \gamma^\kappa \gamma^\mu - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= -\gamma^\kappa \gamma^\mu \gamma^\lambda + 2\gamma^\kappa \eta^{\lambda\mu} + \gamma^\lambda \gamma^\mu \gamma^\kappa - 2\gamma^\lambda \eta^{\kappa\mu} - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= +\gamma^\mu \gamma^\kappa \gamma^\lambda - \gamma^\mu \gamma^\kappa \gamma^\lambda - \gamma^\kappa \gamma^\mu \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa - \gamma^\mu \gamma^\lambda \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} + \gamma^\lambda \gamma^\mu \gamma^\kappa - 2\gamma^\lambda \eta^{\kappa\mu} \\ &\quad - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= +\gamma^\mu \gamma^\kappa \gamma^\lambda - 2\eta^{\kappa\mu} \gamma^\lambda + 2\eta^{\lambda\mu} \gamma^\kappa - \gamma^\mu \gamma^\lambda \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} - 2\gamma^\lambda \eta^{\kappa\mu} - \gamma^\mu \gamma^\kappa \gamma^\lambda + \gamma^\mu \gamma^\lambda \gamma^\kappa \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= -2\eta^{\kappa\mu} \gamma^\lambda + 2\eta^{\lambda\mu} \gamma^\kappa + 2\gamma^\kappa \eta^{\lambda\mu} - 2\gamma^\lambda \eta^{\kappa\mu} \\ 4i [S^{\kappa\lambda}, \gamma^\mu] &= +4\eta^{\lambda\mu} \gamma^\kappa - 4\eta^{\kappa\mu} \gamma^\lambda \end{aligned}$$

We get the required result

$$[S^{\kappa\lambda}, \gamma^\mu] = -i\eta^{\lambda\mu}\gamma^\kappa + i\eta^{\kappa\mu}\gamma^\lambda \quad (6.5)$$

6.1 Part (b)

Further, show that

$$[S^{\kappa\lambda}, S^{\mu\nu}] = i\eta^{\lambda\nu} S^{\kappa\mu} - i\eta^{\lambda\mu} S^{\kappa\nu} + i\eta^{\kappa\mu} S^{\lambda\nu} - i\eta^{\kappa\nu} S^{\lambda\mu} \quad (6.6)$$

$$\begin{aligned}
[S^{\kappa\lambda}, S^{\mu\nu}] &= \left[S^{\kappa\lambda}, \frac{1}{4i} [\gamma^\mu, \gamma^\nu] \right] \\
4i [S^{\kappa\lambda}, S^{\mu\nu}] &= [S^{\kappa\lambda}, \gamma^\mu] \gamma^\nu + \gamma^\mu [S^{\kappa\lambda}, \gamma^\nu] - [S^{\kappa\lambda}, \gamma^\nu] \gamma^\mu - \gamma^\nu [S^{\kappa\lambda}, \gamma^\mu] \\
4i [S^{\kappa\lambda}, S^{\mu\nu}] &= [[S^{\kappa\lambda}, \gamma^\mu], \gamma^\nu] + [\gamma^\mu, [S^{\kappa\lambda}, \gamma^\nu]] \\
4i [S^{\kappa\lambda}, S^{\mu\nu}] &= -i\eta^{\lambda\mu} [\gamma^\kappa, \gamma^\nu] + i\eta^{\kappa\mu} [\gamma^\lambda, \gamma^\nu] - i\eta^{\lambda\mu} [\gamma^\kappa, \gamma^\nu] + i\eta^{\kappa\mu} [\gamma^\nu, \gamma^\lambda] \\
[S^{\kappa\lambda}, S^{\mu\nu}] &= -i\eta^{\lambda\mu} S^{\kappa\nu} + i\eta^{\kappa\mu} S^{\lambda\nu} - i\eta^{\lambda\mu} S^{\kappa\nu} + i\eta^{\kappa\mu} S^{\nu\lambda}.
\end{aligned}$$

check indices

6.2 Part (c)

7 $e^- \mu^- \rightarrow e^- \mu^-$ tree level

7.1 Part (a)

Since the internal line is a photon the propagator is of the form

$$-\frac{ig_{\mu\nu}}{q^2 + i\varepsilon} \quad (7.1)$$

Thus we can read the amplitude to take the form

$$\mathcal{M} = \bar{u}^s(p') [-ie\gamma^\mu] u^s(p) \left[-\frac{ig_{\mu\nu}}{q^2 + i\varepsilon} \right] \bar{u}^s(k') (-ie\gamma^\nu) u^s(k') \quad (7.2)$$

$$\mathcal{M} = -\frac{e^2}{q^2} [\bar{u}^s(p') \gamma^\mu u^s(p)] [\bar{u}^s(k') \gamma_\mu u^s(k)] \quad (7.3)$$

Now since we want to sum over the initial states and average over the final states we need to consider

$$\overline{|\mathcal{M}|^2} = \frac{1}{4} \sum_s |\mathcal{M}|^2 \quad \sum_s u^s \bar{u}^s(p) = \not{p} + m \quad (7.4)$$

$$|\mathcal{M}|^2 = \frac{e^4}{q^4} \bar{u}^s(p') \gamma^\mu u^s(p) \bar{u}^s(p') \gamma^\nu u^s(p) \bar{u}^s(k') \gamma_\mu u^s(k) \bar{u}^s(k') \gamma_\nu u^s(k) \quad (7.5)$$

$$\frac{1}{4} |\mathcal{M}|^2 = \frac{e^4}{4q^4} \text{Tr}[(\not{p}' + m_e) \gamma^\mu (\not{p} + m_e) \gamma^\nu] \text{Tr}[(\not{k}' + m_\mu) \gamma_\mu (\not{k} + m_\mu) \gamma_\nu] \quad (7.6)$$

where

$$(\not{p}' + m_e) \gamma^\mu (\not{p} + m_e) \gamma^\nu = (\gamma^\alpha p'_\alpha + m_e) \gamma^\mu (\gamma^\beta p_\beta + m_e) \gamma^\nu \quad (7.7)$$

A trace of an odd number of gamma's vanishes thus we only need to consider

$$\text{Tr}[\gamma^\alpha p'_\alpha \gamma^\mu \gamma^\beta p_\beta \gamma^\nu] = 4(p^{\mu'} p^\nu + p^{\nu'} p^\mu - g^{\mu\nu} (p \cdot p' - m_e^2)) \quad (7.8)$$

The same relation is true for the k momenta.

$$\frac{1}{4} |\mathcal{M}|^2 = \frac{e^4}{4q^4} 4(p^{\mu'} p^\nu + p^{\nu'} p^\mu - g^{\mu\nu} (p \cdot p' - m_e^2)) 4(k^{\mu'} k^\nu + k^{\nu'} k^\mu - g^{\mu\nu} (k \cdot k' - m_\mu^2)) \quad (7.9)$$

$$\frac{1}{4} |\mathcal{M}|^2 = \frac{16e^4}{4q^4} [2(p' \cdot k') (p \cdot k) + 2(p' \cdot k) (p \cdot k') + 2(p' \cdot p) (k \cdot k') - 2m_\mu^2 p' \cdot p - 2k \cdot k' m_e + 2m_\mu^2 m_e^2].$$

Using Maldenstram Variables

$$s := (p + k)^2 \quad t := (p - p')^2 \quad u := (p - k')^2 \quad (7.10)$$

$$\begin{aligned} p' \cdot k' &= p \cdot k = \frac{s - m_e^2 - m_\mu^2}{2} \\ p \cdot k' &= p' \cdot k = \frac{m_e^2 + m_\mu^2 - u}{2} \\ p' \cdot p &= k' \cdot k = \frac{m_e^2 + m_\mu^2 - t}{2}. \end{aligned}$$

$$\frac{1}{4}|\mathcal{M}|^2 = \frac{32e^2}{16q^4} \left[(s - m_e^2 + m_\mu^2)^2 + (m_e^2 + m_\mu^2 - u)^2 + 2t (m_e^2 + m_\mu^2) \right] \quad (7.11)$$

$$t = q^2$$

$$\frac{1}{4}|\mathcal{M}|^2 = \frac{2e^2}{t^2} \left[(s - m_e^2 + m_\mu^2)^2 + (m_e^2 + m_\mu^2 - u)^2 + 2t (m_e^2 + m_\mu^2) \right] \quad (7.12)$$

7.2 Part (b)

In the limit $\frac{m_e}{m_\mu} \rightarrow 0$ we can drop terms with just m_e

$$\overline{|\mathcal{M}|^2} = \frac{2e^2}{t^2} \left[(s + m_\mu^2)^2 + (m_\mu^2 - u)^2 + 2t (m_\mu^2) \right] \quad (7.13)$$